

1. Project Information

Project Title: Automated Phishing Email Detection & Reporting System

Team Members:

- Aml Mahmoud Ali – 240101441 - Networking & Cybersecurity
- Mazen Ahmed Mamdouh – 240101129 - Networking & Cybersecurity
- El Hassan Waleed – 240101584 – Networking & Cybersecurity
- Omar Ameen El-Sayed - 240102941 – Data Science and AI
- Ahmed Osama - 230105887 - Networking & Cybersecurity

Supervisor: Dr. Taghreed

Program/Department:

Data Science and AI – Networking & Cybersecurity

2. Project Overview

This project focuses on designing and implementing an automated phishing email detection system using machine learning techniques. The system analyzes email content and metadata to classify emails as phishing or legitimate. It emphasizes **data-driven detection using NLP and supervised learning models**, providing a lightweight, privacy-preserving, and fully offline-capable solution.

The final system accepts email text (or extracted email fields), processes it through a trained ML model, and outputs a phishing classification along with confidence scores.

3. Problem Statement & Market Need & Industry collaboration

- **Problem:** Phishing emails remain one of the most effective cyberattack vectors. Many attacks bypass traditional rule-based filters by using well-crafted language, spoofed identities, and subtle social engineering techniques.

- **Market Need:** Organizations and individuals require intelligent, explainable, and cost-effective phishing detection systems that do not depend on paid APIs or external services. ML-based email analysis provides scalable protection suitable for educational, small-business, and enterprise environments.
- **Relevance:** This project aligns with real-world cybersecurity and data science applications, combining NLP, machine learning, and security awareness into a practical system.

4. Objectives

1. Develop a machine learning model to classify phishing and legitimate emails.
2. Extract and engineer meaningful features from email text and headers.
3. Evaluate multiple ML models to improve detection accuracy and robustness.
4. Build a simple backend or script-based interface for email classification.
5. Ensure the solution is privacy-aware, efficient, and reproducible.

5. Innovation & Expected Impact

Innovation:

- Pure ML-based phishing detection without reliance on third-party threat intelligence APIs.
- Use of ensemble and advanced NLP techniques to improve classification performance.
- Focus on explainability through feature importance and evaluation metrics.



Expected Impact:

- Reduced exposure to phishing attacks.
- Improved awareness of phishing patterns and tactics.
- A reusable ML pipeline applicable to other text-based security problems.

6. Methodology & Technical Approach

6.1 Data Collection & Preprocessing

- **Public datasets:** Kaggle phishing datasets, SpamAssassin, Enron email dataset.
- **Email components used:**
 - Subject
 - Body text
 - Optional header-based features
- **Preprocessing steps:**
 - Text cleaning and normalization
 - Tokenization and stopword removal
 - TF-IDF vectorization

6.2 Feature Engineering

- Text-based features (TF-IDF n-grams).
- Linguistic indicators (urgent language, suspicious keywords).
- Structural indicators (URL presence, email length).

6.3 Machine Learning Models

- **Baseline models:**
 - Logistic Regression
 - Naive Bayes
- **Advanced models:**
 - Random Forest
 - Support Vector Machine (SVM)
 - Ensemble / Stacking models
- **Model evaluation using:**
 - Accuracy
 - Precision
 - Recall
 - F1-score

7. Project Deliverables

1. Cleaned and preprocessed phishing email dataset.
2. Feature extraction and preprocessing pipeline.
3. Trained and evaluated ML models.
4. Final phishing detection system (script or backend service).
5. Technical documentation and final report.

8. Timeline (Gantt-style or simple milestones)

Phase 1 – Dataset Preparation (Month 1)

- Collect and clean datasets.
- Perform preprocessing and feature extraction.

Phase 2 – Model Development (Month 2–3)

- Train baseline and advanced ML models.
- Tune hyperparameters and evaluate performance.

Phase 3 – System Integration (Month 4)

- Integrate preprocessing and model into one pipeline.
- Test system on unseen emails.

Phase 4 – Testing & Validation (Month 5)

- Performance benchmarking.
- Error analysis and optimization.

9. Resources Required

- **Hardware: Standard laptops/PCs.**
- **Software & Libraries:**
 - Python
 - scikit-learn
 - NumPy, pandas
 - NLTK / spaCy
 - Matplotlib / Seaborn

10. Risk Assessment & Safety Considerations

Risks:

- Dataset imbalance or bias.
- Overfitting on specific phishing patterns.

Mitigation:

- Cross-validation and proper evaluation.
- Use of multiple datasets and model comparison.

Safety & Privacy:

- Use anonymized and public datasets only.
- No storage of real user emails.

11. References

- Kaggle Phishing Email Datasets
- SpamAssassin Public Corpus
- scikit-learn Documentation
- Research papers on phishing detection using NLP